

Week 1

Lecture 1

1. What is a model?

A model is a simplified representation of a system or phenomenon. The goal is to identify essential characteristics and relationships while leaving out less relevant details (abstraction).

Types of models:

- **Physical:** Tangible objects (e.g., a crash test dummy).
- **Mathematical:** Uses equations and formulas (e.g., $F = ma$).
- **Computational:** Simulates complex systems and dynamics over time.
- **Conceptual:** Organizes ideas into frameworks or categories (e.g., personality types).

2. Why do we model human behaviour?

Modeling gives us the power to predict outcomes, design better systems, and influence positive change.

Applications:

- **Explaining:** How do conspiracy theories spread online?
- **Predicting:** Which video will a specific user like?
- **Experimenting:** Simulating the spread of diseases.
- **Intervening:** Making recommendations that help people make logical decisions.

3. The Scale of Modeling

Behaviour can be viewed on different scales. Choosing the correct scale for your research question is crucial.

- **Micro:** Individual level (e.g., one person).
- **Meso:** Group or organizational level.
- **Macro:** System or aggregated level (e.g., cultural trends).

Important: Laws that apply at the micro-scale often do not work at the macro-scale because of emergent phenomena (complex group patterns that are not visible at the individual level).

4. Risks and “Building Wisely”

A model is not reality; it is always a simplification based on assumptions and limitations in theory or resources. As George Box said:

“All models are wrong, but some are useful.”

How do you build a useful model?

1. **Generate testable hypotheses:** What do you expect to observe if the model is correct?
2. **Validation:** Test the model against new, real-world data.
3. **Generalization:** Does the model also work under different conditions?
4. **Transparency:** Be honest about limitations (e.g., an LLM predicts language use but does not model human thinking).

Implementation: Theory to Model

You can approach a model in two ways:

Theory-first (Inductive)

You start with a theory (e.g., rewards produce dopamine), build a concrete model (e.g., app notifications), and test your hypothesis against data.

Data-first (Deductive)

You collect data, identify patterns (e.g., people click more often on green buttons), capture this in a model, and possibly develop a theory later.

Week 1 — Lecture 2

1. Understanding Human Behaviour (Theoretical Lenses)

To model behaviour, we must first decide from which perspective we view humans. The lecture distinguishes several approaches:

Cognitive Science

Focuses on how the mind works as an information-processing system:
Input → Processing → Output.

Behavioural Sciences

Looks at observable actions and how they are influenced by the environment or rewards (e.g., dopamine release from notifications).

Social Science

Analyzes how individual actions create cascades in markets or social movements (Meso/Macro scale).

2. Theory-driven vs Data-driven Modeling

This is a crucial distinction for your Python and NetLogo assignments.

Theory-first (Inductive)

- Starts from an existing theory (e.g., “people are herd animals”).
- You build a model implementing this theory.
- You test hypotheses against data to refine the theory.

Data-first (Deductive)

- You collect large amounts of data and search for patterns (e.g., correlations in click behaviour).
- You build a predictive model directly from the data (such as a neural network).

Risk: Overfitting

The model works perfectly on the current data but does not understand the underlying logic and fails on new data.

3. The Role of AI in Behaviour Modeling

AI systems are described as “testbeds” for our ideas about how humans think and behave.

LLMs (Language Models)

Model how we speak but are not realistic representations of human cognition.

Recommender Systems

Models predicting what we want based on previous data.

Techno-ethics

Evaluating the impact of these models on society is an essential learning objective.

4. Step-by-step Implementation: Choosing a Restaurant

The slides provide a concrete example of starting a model.

1. Identify components

Determine variables such as:

- price
- distance
- Google rating

2. Identify relationships

Are relationships:

- **Linear?** (the more expensive, the lower the chance of visiting)
- **Non-linear?** (distance matters strongly until 2 km, after that not much)

3. Interaction effects

Does price matter more when someone has a low budget?

Connection with Course Objectives

This lecture lays the foundation for:

- **Objective 1:** Understanding risks (such as limitations of data-driven models).
- **Objective 8:** Synthesizing different perspectives (cognitive vs behavioural).

Extra Reading Material 1

1. What is Behaviourism? (The Basics)

The core assumption of behaviourism is:

“Behaviour is what organisms do.”

The goal is to create a science of behaviour focused only on observable actions rather than invisible internal processes such as thoughts or emotions.

The text distinguishes three forms of behaviourism:

Methodological Behaviourism

A psychological theory about how research should be conducted:
only observe what can be seen.

Psychological Behaviourism

Explains behaviour through external stimuli and past rewards (reinforcement).

Analytical/Logical Behaviourism

Claims that mental concepts (“believing”, “hoping”) are simply descriptions of behaviour or behavioural tendencies.

2. The Role of the Environment (Skinner & Reinforcement)

B.F. Skinner argued that behaviour is shaped by environmental history rather than inner will.

Contingencies

Behaviour is shaped by its consequences.

If an action leads to a positive outcome, the organism is more likely to repeat it.

Anti-mentalism

Thoughts and desires are unnecessary for explaining behaviour.

If you know how the environment rewarded someone in the past, you can predict future behaviour.

3. The “Denial” of the Mind

Behaviourists argue that:

The mind is a by-product

Thoughts are not causes of behaviour but private events occurring alongside behaviour.

No dualism

There is no separation between body and mind.
Everything is physical and follows physical laws.

4. Criticism and the “Zombie Problem”

Phenomenal Qualia

Philosophers argued that pain is not only behaviour (“pulling your hand away”) but also includes the subjective experience of pain.

The Zombie Problem

A being acting exactly like a human without consciousness would still count as human in behaviourism.

Many see this as a weakness.

Complex Inner Life

People may appear inactive while internally fantasizing, analyzing, or theorizing.
Behaviourism struggles to capture this.

Connection with Modelling Human Behaviour

1. Agent-Based Modeling (NetLogo)

Many NetLogo agents are purely behaviourist:
they react directly to their environment.

2. Reinforcement Learning (AI)

Modern AI systems learning through rewards (e.g., AlphaGo) are directly inspired by Skinner’s principles.

3. Critical Perspective

A model focusing only on observable behaviour may predict actions but fail to explain *why* they occur.

Extra Reading Material 2

Where behaviourism focuses on external behaviour, the Cognitive Approach looks inward.

1. The Mind as an Information Processor

The human mind is viewed as an information-processing system similar to a computer.

- **Input:** Information enters through the senses.
 - **Processing:** The brain manipulates this information.
 - **Output:** Behaviour or speech.
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2. Mental Representations and Schemas

Schemas

Internal “maps” or structures helping us understand the world.

Active Processing

Humans actively select, interpret, and distort information based on prior knowledge.

3. Scientific Methods: Indirect Observation

Because thoughts cannot be observed directly, psychologists use indirect methods:

Reaction Times

Longer reaction times indicate more complex mental processing.

Error Analysis

Mistakes reveal how the system works internally.

Computational Models

Building models (e.g., neural networks) to see whether they behave similarly to humans.

4. The Cognitive Revolution and AI

Mutual Influence

Computers helped psychologists understand the mind, while psychology helped AI researchers design smarter algorithms.

Limitations

Data-driven AI may imitate human language but still lacks real human cognition and understanding of the physical world.

Comparison: Behaviourism vs Cognitivism

Aspect	Behaviourism	Cognitive Approach
Focus	Observable behaviour	Internal mental processes
Metaphor	Human as reactive machine	Human as information-processing computer
Thoughts	Ignored or by-products	Central explanation for behaviour
Modeling	Stimulus-response agents	Information processing systems

Application in Projects

Objective 8 (Synthesis)

You must decide whether a problem needs:

- a simple behaviourist model ("if X happens, do Y")
- or cognitive modeling (how users interpret interfaces).

Critical Reflection

Data-driven models are often behaviourist, while theory-first models usually rely on cognitive principles.

Extra Reading Material 3

Social Psychology adds a third dimension:
context.

Human behaviour is strongly influenced by the real or imagined presence of others.

1. The Power of the Situation

Social situations are often stronger than personality traits.

Social Influence

We are constantly influenced by others, even when they are not physically present.

Construction of Reality

Different people may interpret the same situation differently based on background and experiences.

2. Fundamental Needs in Groups

Belonging

Humans strongly desire social connections.

Accurate Worldview

People look to others to determine what is “normal”.

Self-esteem

People try to maintain a positive self-image.

3. Collective Behaviour and Cascades

Small individual actions can create large-scale patterns.

Individual → Macro

Votes, purchases, and social media posts can scale into movements or crises.

Social Networks

This forms the basis for Social Network Analysis (SNA).

Comparison of the Three Perspectives

Perspective	Focus	Metaphor	Role in Course
Behaviourism	Observable actions	Human as reactive machine	Basis for NetLogo
Cognitivism	Information processing	Human as computer	Basis for UI design
Social Psychology	Interaction & groups	Human as network node	Basis for SNA

Ethics and Responsibility

Algorithms on platforms like Facebook or Instagram may unintentionally create polarization and echo chambers by exploiting our need for confirmation and group identity.

Extra Reading Material 4 — The Media Equation

1. What is the Media Equation?

The Media Equation claims:

Media = Real Life

Humans automatically apply social rules to technology.

2. Social Rules for Machines

Politeness

People are more polite to computers evaluating themselves.

Flattery

Compliments from computers affect us positively.

Personality

We assign personalities to devices based on voice and behaviour.

3. Why Does This Happen?

Evolutionary Delay

Our brains evolved in environments where anything speaking or moving was alive.

Automatic Responses

Technology is too new for our brains to stop applying social rules to it.

Connection with the Course

Behavioural Design (Figma)

Interfaces function like “social beings”.

AI Modeling

Humans quickly form emotional bonds with chatbots and robots.

Techno-ethics

Social responses to machines make manipulation easier.

Week 2

1. What is ABM exactly? (The “Upside-Down” Method)

Normally, scientists study the whole system first (for example: “The economy shrinks by 2%”). In Agent-Based Modeling (ABM), you do the opposite: you only program the individuals.

Individual focus

You do not tell the computer what the group should do; you only specify what one individual agent should do.

Autonomy

Each agent makes decisions independently based on its own environment. The computer then calculates what happens when thousands of these agents interact at the same time.

2. The Three Building Blocks in NetLogo

In your practicals, you work with three elements that constantly interact with each other.

Turtles (The agents)

These are the agents in the model.

They have variables such as:

- energy
- color
- heading

Patches (The environment)

The world is made of square cells called patches.

A patch is not just decoration:
it can store information such as:

- amount of grass
- temperature

Observer

The observer is the entity controlling the model and advancing time using the “go” button.

3. How does time work? (Ticks)

Time in ABM progresses in steps called ticks.

Step 1

The clock ticks (tick 1).

Step 2

All agents observe their surroundings and execute their rules.

Example:

“move one step”

Step 3

The world updates.

Step 4

The clock ticks again (tick 2).

This process repeats thousands of times, creating movement and patterns.

4. Emergence: Why do we use ABM?

The main goal of ABM is discovering emergence.

Definition

Emergent behaviour is behaviour that was not directly programmed but appears automatically through interactions.

Classic Example

If birds only follow the rule:

“stay 1 meter away from your neighbour”

a flock naturally emerges.

You programmed the “distance rule”, not the flock itself.

5. Connection with Theory

The Week 1 theories reappear in ABM.

Behaviourist Agents

Simple stimulus-response rules:

if food-nearby? [move-to-food]

Agents only react to stimuli.

Cognitive Agents

More complex agents that require:

- memory
- planning
- internal states

6. The Dangers (Validation and GIGO)

GIGO = Garbage In, Garbage Out

If your rules are unrealistic:

“people always give away all their money”

then your model’s output is also unrealistic.

Validation

You must always compare your simulation results with real-world data.

Understanding Agents in NetLogo

Agents are the “creatures” or “actors” moving around in the virtual world and performing actions.

Think of them as characters in a game following instructions.

The Three Components Explained Again

1. Turtles (Agents)

Turtles can represent:

- people
- birds
- cells
- cars

What can they do?

- move
- change color
- reproduce
- die

Individual Memory

Each turtle can have unique variables:

- age
- energy

Interaction

Agents can sense nearby agents and react.

Example:

“if another turtle gets too close, move away”

2. Patches (Environment)

The NetLogo world consists of square cells called patches.

The Ground

Turtles move across patches.

Not Passive

Patches contain information:

- grass amount
- heat
- resources

Influence on Behaviour

Agents observe the patch they stand on.

Example:

“if the patch is green, eat the grass”

3. Observer

The observer is the programmer or computer running the simulation.

Overview

The observer controls the entire model.

Time Control

The observer advances time through ticks.

Commands

The observer starts the simulation:

- setup
- go

Example: Forest Fire Model

Patches

Trees:

- healthy
- burning
- burned

Agents (Turtles)

Firefighters moving through the forest.

Observer

The user pressing “Start” and observing the spread of fire.

Extra Reading Material

1. Social Systems as “Compositions”

Social phenomena such as:

- traffic jams
- trends
- crowds

are not simply the sum of individuals.

No “Average Person”

Society cannot be understood by examining only the “average citizen”.

Every individual reacts differently.

Connection

Social sciences study how individual actions connect to larger systems.

ABM helps visualize these connections.

2. Micro-Motives and Macro-Behaviour

This is one of the most important concepts.

Micro Level

Individual agent actions and decisions.

Macro Level

Large-scale patterns that emerge.

The Gap

Macro behaviour often surprises us because it does not resemble individual intentions.

Example

Nobody wants to create traffic jams, yet collective braking behaviour produces them.

3. Three Building Blocks in ABM (Social Meaning)

Agents (Turtles)

Social actors.

Heterogeneous

Agents are different from each other:

- age
- opinions
- speed

Interaction Rules

Agents:

- imitate behaviour
- avoid each other
- exchange information

Environment

The physical/social space where interactions occur.

4. Why Computer Simulations Are Necessary

Humans are poor at predicting complex systems.

Non-linear Effects

Small changes can create massive consequences.

Laboratory Function

Since we cannot experiment with real societies easily, ABM acts as a safe testing environment.

Simplified Understanding

Agents (Turtles)

Individuals with social rules.

Patches

Spaces where people move.

Interaction

Moments where turtles influence each other:

- spreading viruses
 - spreading opinions
-

Connection with Course Objectives

Objective 2

Building models in NetLogo.

Objective 6

Critically evaluating simplifications in models.

Extra Reading Material — Joshua Epstein

Joshua Epstein argues that modeling is unavoidable.

1. The Myth of “Model-Free” Science

Everyone models reality, even unconsciously.

Implicit Models

Whenever you predict something:

“higher interest rates lower house prices”

you are using a mental model.

Problem

Mental models:

- are vague
- contain hidden assumptions
- are difficult to test

Explicit Models

Coding models in NetLogo forces all assumptions to become visible.

2. Sixteen Reasons to Model

Models are not only for prediction.

Important purposes include:

Explanation

Understanding why patterns emerge.

Guiding Data Collection

Helping determine what data should be measured.

Discovering “Unknown Unknowns”

Finding surprising outcomes.

Setting Boundaries

Showing what cannot happen under certain conditions.

3. Epstein’s View of the Building Blocks

Agents (Turtles)

Not perfectly rational.

Agents:

- make mistakes
- have emotions
- lack complete information

This is called:

bounded rationality

Interaction Rules

Micro-motives produce macro-behaviour.

Environment (Patches)

The space where local influences matter.

4. Generative Explanation

Epstein's famous quote:

"If you didn't grow it, you didn't explain it."

Meaning

You only truly understand a phenomenon if you can build a model where it naturally emerges.

You must show how turtle interactions on patches create real-world patterns.

Why This Matters for Your Project

This article helps explain:

1. Your model does not need to be perfect to be useful.
 2. Your assumptions about agents.
 3. Why observing macro-behaviour teaches us about reality.
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Extra Video Material

The simulation video perfectly illustrates Epstein's ideas.

Why simulate diseases?

Explanation

Small behavioural changes like handwashing affect entire populations.

Invisible Dangers

Diseases spreading silently are often more dangerous.

Emergence

Random movement patterns create infection curves.

Main Conclusion

Social distancing helps, but:

- early testing
- large-scale testing
- isolation

are the most effective strategies during pandemics.

Lecture 2 — Basics of NetLogo

This lecture focuses on building scientifically reliable ABMs.

1. The Anatomy of an Agent

An agent contains three major components.

State

Internal variables:

- color
- energy
- wealth

Behaviour

Rules determining state changes.

Example:

“if energy < 10, search for food”

Memory

Some agents store past experiences and adapt future behaviour.

2. Types of Interactions

Direct Interaction

Agent A directly affects Agent B:

- infection
- fighting
- mating

Indirect Interaction (Stigmergy)

Agents communicate through patches.

Example:

- ants leaving pheromone trails

Network-Based Interaction

Agents only interact through network links.

3. Environment: Space and Structure

Patches define context.

Topology

The world may be:

- flat
- torus-shaped
- network-based

Torus

Leaving one edge causes re-entry from the opposite edge.

Resources

Patches provide:

- energy
 - information
 - survival resources
-

4. Validation: Is Your Model “True”?

Face Validity

Does behaviour appear logical to experts?

Empirical Validation

Do macro patterns match real-world data?

Sensitivity Analysis

Do small parameter changes drastically alter outcomes?

5. Conceptual Design: ODD Protocol

ODD stands for:

- Overview
- Design Concepts
- Details

It forces you to define:

Goal

What phenomenon are you explaining?

Entities

Who are the turtles and patches?

Processes

What happens each tick?

Practical Connection

When stuck in your assignment, ask yourself:

- What state variables does my turtle need?
- Is interaction direct or patch-based?
- How will I validate reliability?

Week 3

Lecture 1 — Social Network Analysis

1. What is Social Network Analysis (SNA)?

SNA is a method for studying social structures using networks and graph theory. The main idea is that someone's position in a network influences their opportunities, power, and access to information.

Basic elements

Concept	Meaning
Nodes	The actors in the network, such as people, organisations, or webpages
Edges	The relationships between actors, such as friendships, communication, or transactions

2. Network Metrics: Measuring Importance

To analyse a network, we use mathematical measures called **centralities**.

Degree Centrality

Measures how many direct connections a node has.

This shows who is the most popular or has the most contacts.

Betweenness Centrality

Measures how often a node lies on the shortest path between other nodes.

This shows who acts as a **bridge** or **gatekeeper** for information.

Closeness Centrality

Measures how quickly a node can reach all other nodes in the network.

This shows who can spread information quickly.

3. The “Small World” and Network Structures

Clusters

Groups of people who are strongly connected to each other, such as friend groups.

Six Degrees of Separation

The idea that everyone in the world is connected through a short chain of around six steps.

Homophily

The tendency of people to connect with others who are similar to themselves.

In simple words:

“birds of a feather flock together.”

4. Why do we model networks?

Network models help us understand social problems at **meso** and **macro** levels.

Information spread

How do viral videos or fake news spread?

Disease spread

SNA helps identify **super-spreaders** in epidemics.

Resilience

What happens if an important bridge person is removed from the network?

Practical Connection: Python

In this week's practical, you implement these ideas in Python.

Concept	Python equivalent
Add a node	<code>G.add_node("Alice")</code>
Create a relationship	<code>G.add_edge("Alice", "Bob")</code>
Find important people	<code>nx.degree_centrality(G)</code>

Critical Reflection

A network model is always a simplification.

For example, a line between two people in Python does not mean they truly trust each other. It only means the data shows some type of connection.

Extra Reading: Barabási-Albert Model

The Barabási-Albert model explains why many real-world networks have a few very large **hubs** and many small nodes.

This is called a **scale-free network**.

Lecture 2 — The Barabási-Albert Model

1. Why the BA Model?

Older network models assumed that connections form randomly. The BA model argues this is unrealistic for social systems.

It is based on two mechanisms:

Growth

Networks are not static. New nodes constantly enter the network.

Preferential Attachment

New nodes prefer to connect to nodes that already have many connections.

This is also called:

“the rich get richer.”

2. How do hubs emerge?

Imagine a new student joining a class. According to the BA model, this student is more likely to become friends with the popular student than with someone isolated.

Hubs

Nodes that entered early and gained many connections can grow into large hubs.

Power Law

A small number of nodes get most of the connections, while most nodes have very few connections.

3. Mathematical Properties

Average Degree

In a standard BA model, the average degree is:

$$\left[\frac{2m}{n} \right]$$

where **m** is the number of edges each new node brings.

Degree Distribution

The degree distribution follows a **power law** with an exponent of 3.

This means very highly connected nodes are rare, but still much more common than in a random network.

4. Limitations of the BA Model

The BA model is useful, but it is still a simplified model.

No deletion

Nodes and links never disappear.

In reality:

- websites disappear
- friendships end
- accounts become inactive

No intrinsic quality

The model mainly looks at who arrived early.

It does not include **fitness**, meaning that a new node might grow quickly because it is simply better or more attractive.

Connection with Modelling Human Behaviour

Micro-Macro Link

A simple individual rule, preferential attachment, creates a large network structure.

Social Network Analysis

This explains why Python assignments often reveal a few nodes with very high degree centrality.

Ethics

Scale-free networks are vulnerable to targeted attacks on hubs.

If the most important person in a network is removed, communication can collapse.

Lecture 2 — Small World, Centrality, and Communities

1. The Small-World Phenomenon

Large real-world networks, such as the internet or friendship networks, are surprisingly “small”.

Short paths

The average distance between two random nodes is small compared to the total size of the network.

Logarithmic growth

Even if a network has millions of nodes, path lengths grow slowly.

2. Advanced Centrality Measures

Betweenness Centrality

Measures how often a node lies on the shortest path between others.

People with high betweenness are:

- bridges
- gatekeepers
- connectors between groups

If they are removed, the network may split into separate parts.

Closeness Centrality

Measures how close a node is to all others.

This shows how quickly information can spread from that node.

3. Meso Scale: Community Detection

Networks often contain groups where nodes are strongly connected internally but weakly connected to the outside.

Modularity

A measure of how well a network is divided into groups.

High modularity means the groups are clearly separated.

Louvain Algorithm

A popular method for finding communities.

It works by:

1. placing nodes into neighbouring groups to increase modularity
 2. merging groups into a smaller network to detect larger structures
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4. Beyond Simple Graphs

Real-world relationships are not always equal.

Directed graphs

Relationships have direction.

Example:

A follows B on Instagram, but B does not follow A.

Weighted graphs

Edges have values.

Example:

how often two people call each other.

Temporal graphs

Relationships change over time.

Some connections only exist at specific moments.

Practical Connection: Python and NetworkX

Concept	Python command
Find bridge people	<code>nx.betweenness centrality(G)</code>
Measure information speed	<code>nx.closeness centrality(G)</code>
Find groups	<code>nx.community.louvain_communities(G)</code>

Why is this important for human behaviour?

Meso-structure

Communities help identify subcultures or echo chambers.

Interventions

If you know who has high betweenness, you know who can spread a trend or help stop a virus.

Extra Reading: Triadic Closure

1. What is Triadic Closure?

Triadic closure means that if two people share a mutual friend, they are more likely to become friends themselves.

Example:

If **B** and **C** are both friends with **A**, then B and C may later form a connection too.

This creates a triangle in the network.

2. Why does this happen?

Opportunity

If A meets both B and C, then B and C may meet through A.

Trust

Knowing that someone is a friend of your friend creates a basic level of trust.

Social Pressure

It may feel awkward if two close friends of A do not know or like each other, so pressure exists to connect them.

3. Strong and Weak Ties

Tightly-knit groups

Groups where almost everyone knows each other.

These groups often contain repeated information and similar opinions.

Bridges

Connections that link two different social worlds.

Local Bridges

In real networks, perfect bridges are rare. Local bridges provide shorter paths to other parts of the network.

4. The Value of Weak Ties

Access to new information

Strong ties often know the same things as you.

Weak ties can give access to:

- new ideas
- new jobs
- new social circles

Structural importance

A bridge tie is important because removing it can isolate parts of the network.

Connection with Modelling Human Behaviour

Clustering Coefficient

Triadic closure explains why social networks often have many triangles.

SNA Metric

People with high betweenness centrality are valuable because they connect separated groups.

Emergence

A micro-mechanism, friends of friends meeting, creates macro-structures of dense clusters connected by weak bridges.

Extra Reading: Heterogeneity and Centrality

1. Heterogeneity in Networks

Many networks are heterogeneous.

Hubs

A small number of nodes have extremely many connections.

Examples:

- major airports
- Google
- influencers

The Mass

Most nodes have only a few connections.

2. Centrality Measures

Degree Centrality

The simplest measure.

It counts direct neighbours.

Betweenness Centrality

Measures how often a node lies on shortest paths between other nodes.

Bridges

Nodes with high betweenness connect different network parts.

Communities

Betweenness can also apply to edges.

Edges with high betweenness often connect dense communities.

3. Normalisation

Centrality values depend on network size.

To compare different networks, values must be normalised.

For betweenness, this means dividing the raw score by the maximum possible number of paths through a node.

4. Python Implementation

Metric	Python command
Node betweenness	<code>nx.betweenness_centrality(G)</code>
Edge betweenness	<code>nx.edge_betweenness_centrality(G)</code>

Connection with Modelling Human Behaviour

Micro-Macro Link

A node's individual position affects the structure and resilience of the whole system.

Identifying Groups

Removing high-betweenness links can help isolate communities.

Social Influence

People with high betweenness often have valuable weak ties because they bridge different social worlds.

Extra Reading: Communities

1. What is a Community?

A community is a group of nodes with many internal connections and fewer connections to the outside.

Internal density

Many connections inside the community.

External scarcity

Only a few bridges connect the group to others.

2. Modularity

Modularity, written as **Q**, measures the quality of a community division.

It compares:

- the number of edges inside groups
- with the number expected if edges were random

Meaning

A high Q value means the community structure is strong and unlikely to be random.

In social networks, a good Q value is often between:

[
0.3 \text{ and } 0.7
]

3. Community Detection

Hierarchical Clustering

A method similar to a family tree.

Nodes are grouped step by step based on similarity.

Louvain Algorithm

A fast and popular algorithm.

It tries to maximise modularity by grouping nodes intelligently in phases.

4. Overlapping Communities

In real life, people belong to several groups at once.

Example:

- family
- sports team
- university

Node Membership

A node can belong to multiple communities.

Clique Percolation

A method for finding overlapping groups by looking at small fully connected groups, called cliques.

Course Connection

Micro-Macro Link

Individual choices about who to connect with can lead to a society full of subcultures and echo chambers.

SNA Practice

NetworkX can be used to visualise groups and answer questions like:

Are the freshers and seniors really separated, or do they mix?

Ethics

Community detection can be used for exclusion or targeted manipulation, such as political ads aimed at one specific cluster.

Week 4

Lecture 1 — Behavioural Design and Dark Patterns

In Week 4, the focus shifts from **understanding behaviour** through ABM and SNA to **influencing behaviour** through technology. This lecture forms the basis for the UI design project in Figma and discusses the ethical boundary between helpful design and manipulation.

1. From Models to Design Patterns

In the previous weeks, you learned how behaviour emerges. Now, you use that knowledge to intervene.

Strategising and intervening

Models help us determine the conditions needed to achieve desired behaviour.

Choice architecture

The way information is presented influences the choices people make.

Examples:

- the order of options in a menu
 - the default setting
 - how visible or hidden certain choices are
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2. Psychological Heuristics in Design

Designers use mental shortcuts, or **heuristics**, from the cognitive approach.

Social Proof

People look at others to decide what good behaviour is.

Example:

“80% of users chose this plan.”

This connects directly to social psychology.

Scarcity

Scarcity creates urgency.

Example:

“Only 2 items left in stock!”

Authority

People trust information faster when it comes from an expert or authority figure.

3. Dark Patterns: The Shadow Side

Dark patterns are interfaces designed to push users into actions they did not actually want to take, such as subscribing or sharing more data.

Roach Motel

Easy to enter, difficult to leave.

Example:

signing up is easy, but cancelling is difficult.

Sneak into Basket

An extra product or insurance is added to the basket without clear consent.

Misdirection

The interface distracts the user from important information.

Example:

a large shiny “Yes” button and a nearly invisible “No” option.

4. Ethics and the Media Equation

The theory of **The Media Equation** by Reeves and Nass returns here.

Because people treat computers as social actors, they may feel:

- rude
- guilty
- emotionally attached

towards an interface.

Apps like Woebot use this social response to build a bond. The ethical question is:

Where does care stop, and where does exploitation begin?

5. Looking Ahead: Figma and the Project

Figma

This week, you learn how to make high-fidelity prototypes.

Redesign

A key part of the practical is identifying dark patterns in an existing app and redesigning them into a more ethical, human-centered design.

Connection with Course Objectives

This lecture is essential for:

Objective 5

Identifying ethical risks in AI and behavioural models.

Objective 7

Translating cognitive and social theories into a concrete interface design.

Objective 8

Combining:

- behaviourist rules
 - cognitive information processing
 - social pressure
 - human reactions to machines through the Media Equation
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Shorter Summary Version

1. From Models to Design Patterns

You learn how theoretical knowledge about behaviour can be used to intervene in user choices.

This is also called **choice architecture**: the way options are presented influences what people choose.

Example:

setting one option as the default.

2. Psychological Heuristics

Designers use mental shortcuts to guide users.

Social Proof

“Others also bought this.”

Scarcity

“Only 3 items available!”

Authority

Using experts or well-known logos to create trust.

3. Dark Patterns and Ethics

A major part of this lecture is about the negative side of design: interfaces that mislead users.

Dark Patterns

Example:

Roach Motel: easy to sign up, difficult to cancel.

Ethics

The key question is where helpful design ends and manipulation begins, especially because of the Media Equation: people treat computers like social beings.

4. Figma Project

This lecture prepares you for the assignment where you must:

1. find an existing bad or unethical design
 2. explain why it is unethical
 3. redesign it in Figma into a human-centered solution
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Lecture 2 — Figma and Prototyping

1. What are we actually building? The hierarchy

Before starting, it is important to understand the difference between three stages of design. The assignment often asks for a **high-fidelity prototype**.

Wireframe

A simple blueprint.

It has:

- no colour
- no images
- only layout and functions

Mockup

A “photo” of the app.

It includes:

- colours
- fonts
- images

But it is not clickable.

Prototype

An interactive simulation.

It both:

- looks like the real app
 - behaves like the real app
-

2. Figma Fundamentals

Frames

Frames are your screens.

Everything you design should be inside a frame so it can later be presented properly.

Layers

Like Photoshop, elements are stacked on top of each other.

Components

Reusable building blocks, such as buttons.

If you edit the main component, all copies update automatically.

This saves a lot of time.

Auto Layout

A powerful feature that lets your design automatically adjust to the amount of text.

Example:

a button becomes wider when the text is longer.

3. Interactivity: Creating the Flow

The most important part of this lecture is moving from the **Design** tab to the **Prototype** tab in Figma.

Connections

You draw lines from a button on Screen A to Screen B.

Interactions

You decide how the transition happens.

Examples:

- On Click
- On Hover
- While Pressing

Animations

With **Smart Animate**, Figma recognizes elements across two screens and smoothly animates them between positions.

4. Connection with the Assignment: Redesign and Ethics

In this week's practical, you apply the theory about heuristics and dark patterns.

1. Identify

Find an existing dark pattern.

Example:

a confusing cancellation button.

2. Critique

Explain why it is unethical based on theory.

3. Redesign

Build a new version in Figma that is human-centered.

This means the design helps the user instead of misleading them.

Practical Tips for the Figma Project

Use UI Kits

You do not have to draw everything yourself.

You can download existing iOS or Android libraries in Figma to use real buttons and icons.

Constraints

Pay attention to how elements are fixed:

- left
- right
- centre

This determines whether your design breaks when the frame size changes.

Week 5

Lecture 1 — Data-Driven Models

In this lecture, there is a major shift in how we model human behaviour: from **explicit models** to **implicit, data-driven models**.

1. The Shift from “Rules” to “Data”

Until now, with **ABM** and **SNA**, you worked with explicit models. This means you, as the researcher, define the rules yourself.

Example:

agents move randomly

You understand the mechanism because you built it yourself.

Now, we move to **implicit models**. Here, you do not give the computer rules. Instead, you give it large amounts of data, and the computer learns patterns from that data.

Advantage

Very high predictive power, often better than human experts.

Disadvantage

They are often **black boxes**. You can see what the model predicts, but it is difficult to understand why.

This connects to Joshua Epstein’s distinction between **prediction** and **explanation**.

2. Machine Learning as a Tool

Machine learning is divided into two main categories.

Supervised Learning

The algorithm is trained on data where the correct outcome is already known.

Goal

To predict:

- a **label**, called classification
Example: is this behaviour fraud or not?
 - a **number**, called regression
Example: how much money will this customer spend?
-

Unsupervised Learning

The algorithm receives data without labels and must discover structure by itself.

Goal

Clustering.

Example:

grouping citizens based on online search behaviour

3. Digital Footprints and Personality: OCEAN

A central theme is that our daily use of technology creates data that can reveal psychological traits.

OCEAN / Big Five personality model

Letter	Trait	Meaning
O	Openness	openness to new experiences
C	Conscientiousness	being organised and responsible
E	Extraversion	being outgoing and social
A	Agreeableness	being cooperative and kind
N	Neuroticism	emotional instability or sensitivity

4. The Kosinski Study

Research showed that algorithms could use Facebook likes to estimate someone's personality more accurately than:

- colleagues
- friends
- partners

This shows that data-driven models can detect patterns that are invisible to humans.

5. Psychographic Targeting

Companies and political campaigns can use personality profiles for **micro-targeting**.

Instead of one general advertisement, each person receives a message adapted to their personality.

Example:

a fear-based advertisement for someone high in neuroticism

6. Ethics: The Shadow Side of Predictive Models

The lecture warns about using data-driven models for decisions that strongly affect people's lives.

Case: COMPAS

COMPAS is an algorithm used in the United States to predict the risk of reoffending.

Bias problem

Because the model is trained on historical data, it can reproduce existing human biases.

Research by ProPublica showed that COMPAS gave more **false positives** to certain groups.

Proprietary models

Another problem is that many algorithms are company-owned and secret.

This makes it difficult for:

- lawyers
- citizens
- researchers

to check whether decisions are fair.

7. Main Exam Point

The key trade-off is:

interpretability vs accuracy

Explicit models

Example: NetLogo.

They are:

- easier to understand
- simpler
- more transparent

Implicit models

Example: machine learning.

They are:

- powerful at prediction
 - harder to explain
 - ethically risky
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Expanded Lecture 1 Summary

This lecture marks an important shift in the course. We move from explicit models, such as agent-based modelling and network analysis, where we program assumptions about behaviour ourselves, to data-driven implicit models, where structure is derived from behavioural data.

Although these models often predict very well, their mechanisms are harder to explain.

1. Predictability and Uncertainty

Human behaviour may seem chaotic, but it is not completely random.

Physical mobility

Mobile phone data from 50,000 users showed that people's movement patterns are highly regular.

The theoretical upper limit for predicting someone's next location is about:

93%

Online behaviour

Web browsing also follows routines.

The upper limit for predicting the next website someone visits is around:

85%

Uniqueness

Even though people are predictable, they are also unique.

Only 4 time-location points can identify 95% of people in a database of 1.5 million mobile users.

Also, 4 credit-card transactions are often enough to identify someone.

2. Types of Data

Data-driven models require understanding what type of data is being used.

Structured data

Data with a fixed model, such as SQL tables.

It is easy to analyse and understand.

Unstructured data

Data without a fixed schema.

Examples:

- videos
- images
- social media posts

Around 80% to 90% of organisational data is unstructured.

It contains rich information but needs processing before it can be used.

Semi-structured data

Data with tags or markers but without a strict relational model.

3. Machine Learning as a Tool

Machine learning is a subset of AI that allows systems to learn from experience without being explicitly programmed.

Supervised Learning

The algorithm learns a mapping function from input $\mathbf{x}$ to labelled output $\mathbf{y}$.

Regression

Predicting a quantitative numerical value.

Example:

income or hours of sleep

Classification

Predicting a qualitative category.

Example:

marital status or type of flower

Unsupervised Learning

The algorithm discovers structure in data without known labels or outcomes.

Clustering

Grouping observations into similar subsets.

Dimensionality Reduction

Representing complex data using fewer variables while keeping the most important information.

4. Case Study: Personality and Digital Footprints

A large part of the lecture focuses on predicting the Big Five personality traits:

- Openness
- Conscientiousness
- Extroversion
- Agreeableness
- Neuroticism

Facebook Likes

In the famous study by Kosinski et al., Facebook likes from 58,000 users were used to predict personality.

The model could also predict:

- political preferences
- sexual orientation

Computer vs Human

Follow-up research showed that computer models based on likes could often estimate personality better than:

- friends
- family
- partners

depending on the number of likes available.

5. Influence and Decision-Making

When personality is known, it can be used for **psychographic targeting**.

This means adapting messages to psychological profiles to influence behaviour at scale.

Effectiveness

Field experiments showed that personality-matched advertisements produced up to:

- 40% more clicks
- 50% more purchases

Cambridge Analytica

Cambridge Analytica used these techniques politically by building profiles of 87 million Americans without consent and targeting them with personalised political ads.

Decision-making: COMPAS

The lecture ends with a warning about using data models for life-changing decisions.

COMPAS is a black-box proprietary model used in the United States to predict reoffending risk.

It uses 137 secret factors, raising serious concerns about transparency and fairness.

Lecture 2 — Explainable AI

In the second lecture of Week 5, the focus shifts to a key question:

If data-driven models are powerful but often black boxes, how can we still understand them?

This lecture focuses on **Explainable AI**, also called **XAI**, and the balance between predictive performance and explainability.

1. The Black Box Problem and the Trade-off

The main challenge of modern AI is the tension between performance and understandability.

Interpretability vs Accuracy

There is often an inverse relationship.

Simple models

Examples:

- linear regression
- decision trees

They are:

- easy to understand
- interpretable

But they may not capture complex human patterns.

Complex models

Examples:

- deep learning
- random forests

They are:

- highly accurate
- difficult to understand

It is often impossible to see directly why they make a specific decision.

2. Why do we need explanations?

Explanations are not just for curiosity.

We need them for:

Trust

Users accept decisions more easily when reasons are provided.

Example:

a medical diagnosis with an explanation

Ethics and Justice

Explanations help check whether a model discriminates due to hidden bias in the data.

Law

The GDPR gives citizens, in many cases, a “right to explanation” for automated decisions.

3. Methods to Open the Black Box

Because we often cannot directly read the internal mathematics of complex models, we use **post-hoc techniques**.

Post-hoc means:

explanation after the prediction has already been made

The two most famous methods are **LIME** and **SHAP**.

LIME

LIME stands for:

Local Interpretable Model-agnostic Explanations

LIME does not explain the whole model. It focuses on one specific prediction.

How it works

It takes one datapoint, for example:

a person whose loan application was rejected

Then it slightly changes the input data to see how the prediction changes.

This is called **perturbation**.

Result

LIME builds a simple local model around that one case.

Example explanation:

For this person, the loan was mainly rejected because of low income, even though other factors also played a role.

SHAP

SHAP stands for:

SHapley Additive exPlanations

SHAP is based on game theory.

It distributes the prediction result across the input variables, called features.

Core idea

It calculates the contribution of each variable by comparing predictions with and without that variable across different combinations.

Advantage

SHAP is mathematically strong and can explain:

- individual predictions
- the overall model

It can show which features are most important.

4. Counterfactual Explanations: The “What if?” Method

A very human-friendly form of explanation is the **counterfactual explanation**.

Definition

Instead of explaining the whole model, a counterfactual tells you what would need to change to get a different outcome.

Example

If your income had been €500 higher per month, your loan would have been approved.

Strength

This is useful because it gives the user a clear action point.

They know what could be changed to get a different result in the future.

5. Human-in-the-Loop and Accountability

The lecture ends with the conclusion that AI models should support human decision-making, not replace it completely.

Transparency

We need to understand when and why a model fails, for example due to bias.

Responsibility

Final responsibility for decisions in areas like law or healthcare should remain with humans.

Humans should use the model's explanation as one part of a broader decision.

Summary for Understanding

In Lecture 1, you learned that data can reveal a lot about us.

In Lecture 2, you learn that this power must be controlled through explanation.

Whether we use:

- **LIME** for a quick local explanation
- **SHAP** for fair feature contribution

- **counterfactuals** to understand how to change an outcome

the goal is to turn AI from an incomprehensible “god” into a controllable tool.